

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Detailed Design Document

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niri_h ty_002 / 09 / 1999-10-23

This report gives a detailed software design for the Gemini “Near Infra-Red Imager” and On Instrument Wavefront Sensor.

1 Introduction

This report presents a the software design for the control software for the Gemini Near Infra-Red Imager.

This report supersedes the preliminary document, niri_h ty_000.fm, “Optical-Components Controller: Implementation Issues,” which was released as part of the NIRI PDR documents.

Note that the design of many of the hardware elements is not finalized at this time. Some control algorithms may need to be modified.

1.1 Purpose of the document

This report gives a describes the differences between the design of the software for the Near Infra-Red Imager (NIRI) and the prototype instrument which is described in the CICS documents [14]. This document includes a description of the software for both NIRI and the NIRI OIWFS¹ components controllers. At some time, the OIWFS documents will be placed into a separate document.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS);
- the developers of the Gemini Acquisition and Guidance units;

1. For purposes of this document, the terms NIRI OIWFS and OIWFS will be considered synonymous, though the NIRI OIWFS will be used by other systems.

- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS.

It is assumed that readers of this document are familiar with the CICS prototype instrument system which is described in [14], [15], [16]. Throughout this document, unless otherwise stated, it is assumed that the software design for the NIRI control software is the same as the software design for the CICS prototype instrument

1.2 Scope

The software product to be developed from this document will be the control system for the Near Infra-Red Imager (NIRI).

1.3 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.4 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.5 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Implementation

2.1 Overview

The NIRI and NIRI OIWFS control software will be based on the Core Instrument Control System (CICS). Whenever possible, CICS commands will be duplicated. New commands and their corresponding records will be as similar to existing CICS commands as possible. If there is nothing in the CICS which can be adapted to our purposes, a new command will be designed and implemented, using the specifications in the ICD’s (especially ICD 7a and ICD 1b).

The instrument sequencer (IS) will forward sequence commands to the components controller (CC) and to the detector controller (DC); it will get the World Coordinate System (WCS) information from the TCS; it will modify the WCS information to include instrument-specific information; it will forward the modified coordinates to the DC; and it will set the TCS pointing origin.

The detector controller is a separate work package, and is described in [18].

FIGURE 1. Module Dependencies

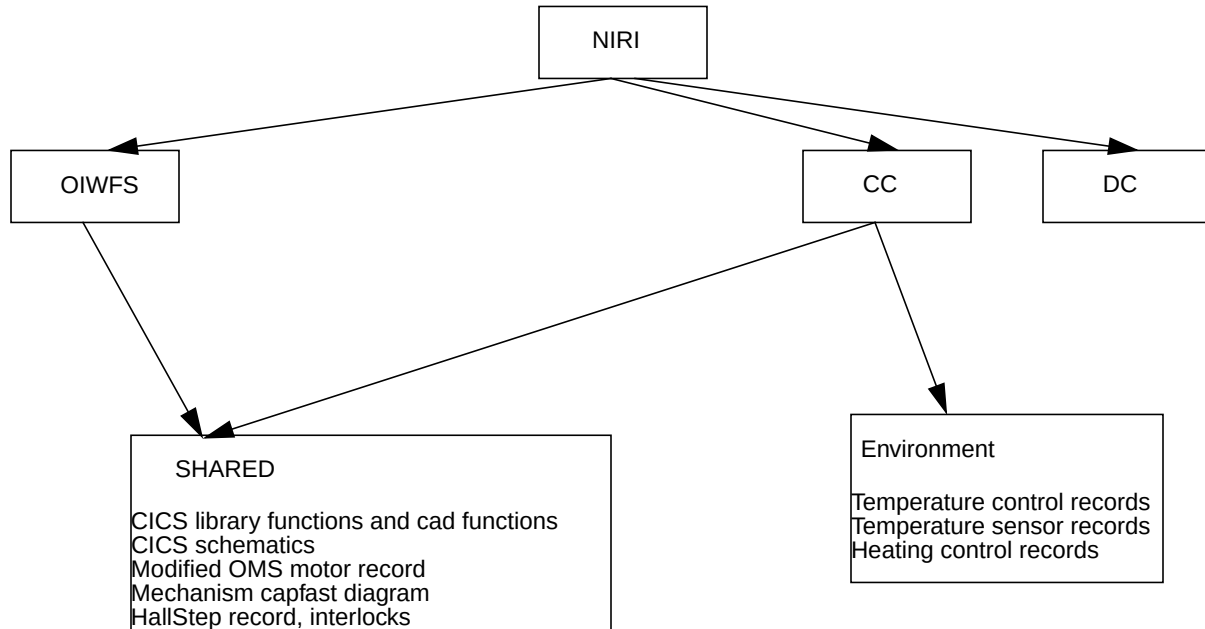
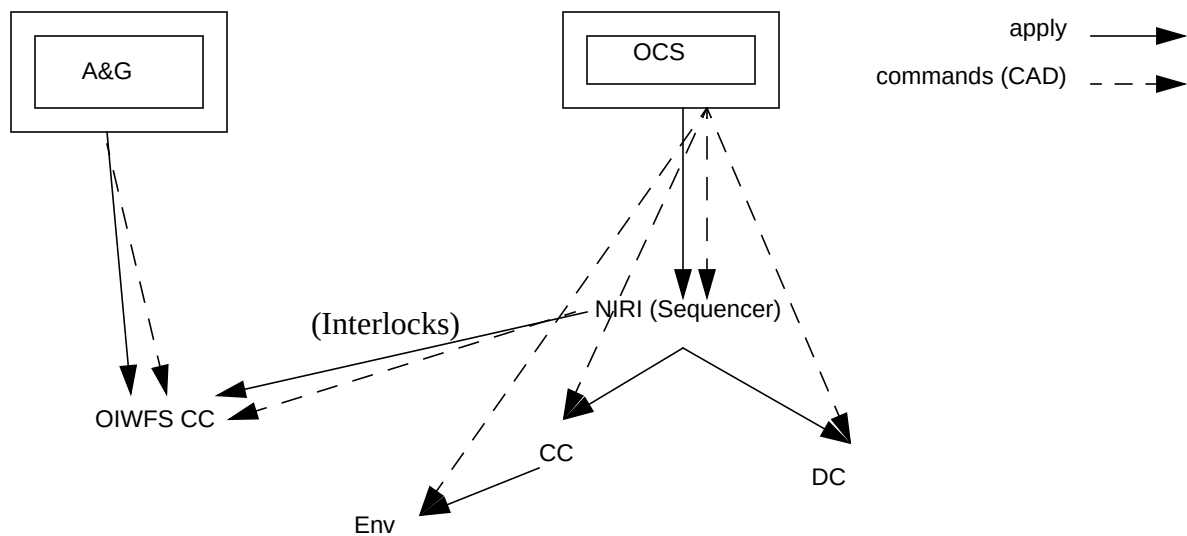


FIGURE 2. Command Flow



2.2 Static Storage

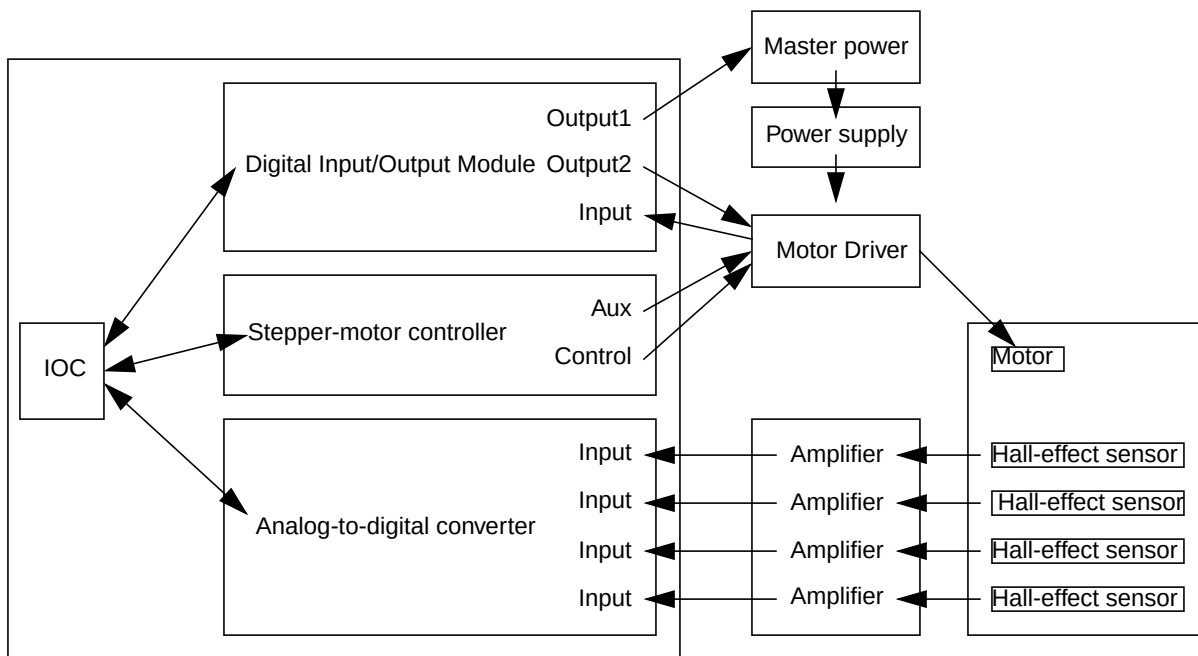
2.2.1 Static Look-Up Tables

Upon initialization of the system, calibration information will be loaded into appropriate EPICS process variables using pload. This information will be necessary to calibrate the Hall-effect sensors. The calibration data will be generated by a separate program, which will be described in [5].

3 Stepper Motor / Hall-Effect Sensor Control

All mechanisms will be controlled using a combination of stepper motors and Hall-effect sensors.

FIGURE 3. Control system overview (single mechanism)



3.0.1 Notes on Figure 3

- All inputs in Figure 3 lead in to the same amplifier and A/D boards. This will probably not be true. The inputs may be distributed among several boards, so that the system will remain partially functional if a single board fails.
- Although the master power control is shown in this single-component diagram, this will actually be shared among all mechanical components. The master power will be shut off only when all components are idle.
- The Digital Input/Output, “Output2” and “Input” are used to check the status of the stepper motor driver, and to reset it if there is a fault.

- The Stepper-motor controller Aux output will be used to shut down the power to the motor when it is not in motion.

3.1 Hall-Effect Sensors and the Home Position

The stepper motors are not encoded. Instead, each component will have a magnet at its datum (home) position. A Hall-effect sensor will be used to detect this magnet to locate the datum position. For some components there may also be other magnets giving physical limits and other reference positions. These secondary magnets may be distinguished from the home-position magnet by using the opposite polarity, and may be distinguished from each other by a different peak field strengths at the sensor².

One sensor is designated as the “primary” sensor. This sensor will be connected to the “home” input of the Oregon Microsystems stepper-motor controller, which will speed up the datuming process. If this sensor fails, then the datum position may still be found, but will require a slower search process.

3.1.1 A discrete positioning mechanism (4-position filter wheel)

Figure 4 describes a simple, 4-position filter wheel. For mechanisms with discrete positions (e.g., filter wheels) there will be reference magnets for each distinct position, as well as the magnets for the home position. For the filter wheel, the reference magnets (and their corresponding sensors) are mounted on two concentric rings. By using a combination of magnets with different field strengths at the sensor, each position may be uniquely encoded. Note that if either sensor fails, the system will continue to operate, though the encoding scheme may no longer be unique.

2. Different field strengths at the sensor may be produced either by varying the depth of the magnet or by using magnets with different strengths.

FIGURE 4.

A simple mechanism (wheel)

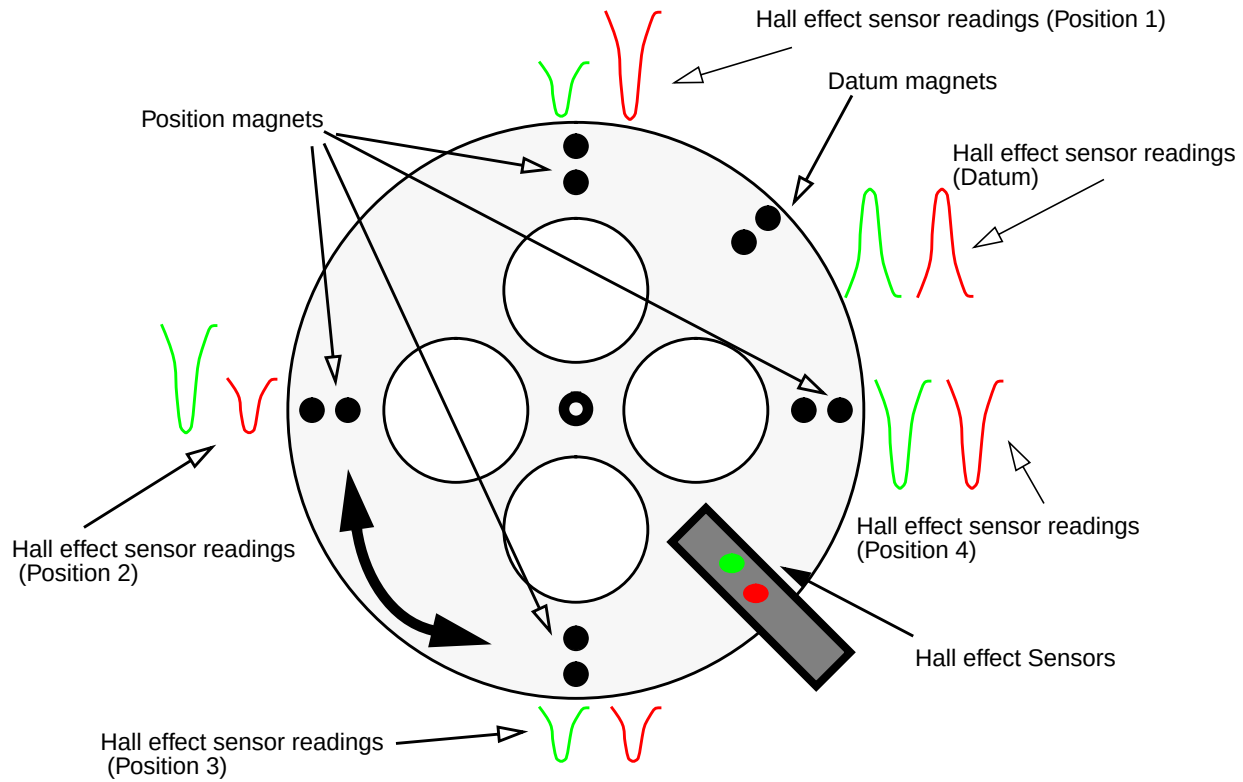


Table 1 describes the sensor values corresponding to Figure 4.

TABLE 1.

Sensor Values for Figure 4

Position	Sensor 1 (V)	Sensor 2 (V)
1	-2.0	-4.0
2	-4.0	-2.0
3	-2.0	-2.0
4	-4.0	-4.0
Datum	+4.0	+4.0

Table 2 describes typical sensor values for a twelve-position wheel. Note that it is assumed that it is only possible to distinguish four different magnet strengths. In practice, it may be possible to distinguish more values.

TABLE 2.

Typical Sensor Values (12 position wheel)

Position	Sensor 1 (V)	Sensor 2 (V)
1	-4.0	-4.0
2	-4.0	-3.0
3	-4.0	-2.0
4	-4.0	-1.0
5	-3.0	-4.0
6	-3.0	-3.0
7	-3.0	-2.0
8	-3.0	-1.0
9	-2.0	-4.0
10	-2.0	-3.0
11	-2.0	-2.0
12	-2.0	-1.0
Datum	+4.0	+4.0

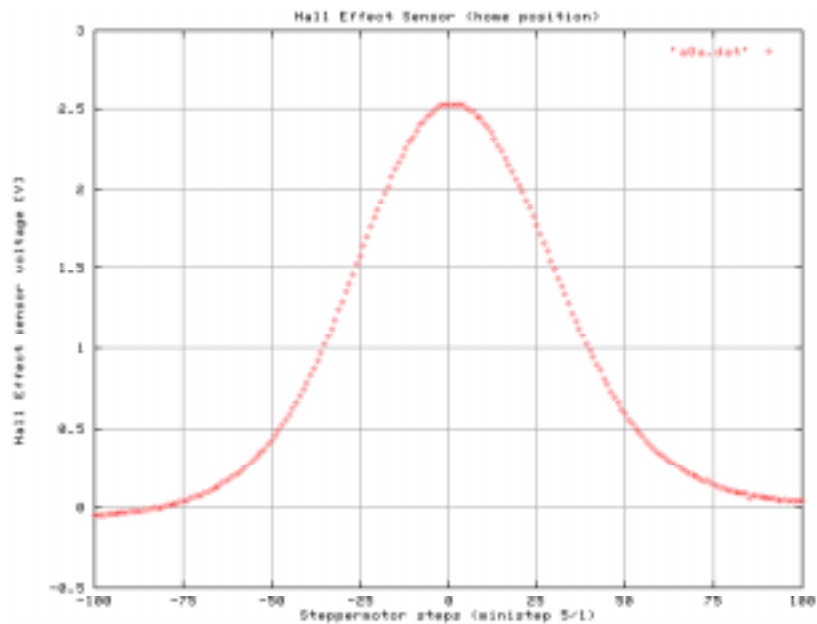
For smaller mechanisms with a small number of positions, the two-sensor arrangement will be modified to use a single back-to-back sensor pair.

3.1.1.1 Sensor data plots (home magnet)

Figure 5 is a data plot of the Hall-Effect sensor values near a datum magnet. Note that the peak is sharply defined and nearly symmetric, though careful inspection shows a slight asymmetry. The data values become negative in the wings of the peak. This is probably a real effect, due to the reversal of the direction of the magnetic field near a magnet.

This data was taken using a 5/1 mini-step stepper motor controller. In practice, the mini-steps may prove to be useless, because the motor will tend to fall into a detente when power to the motor is removed.

FIGURE 5. Hall effect sensor readings (Datum magnet)



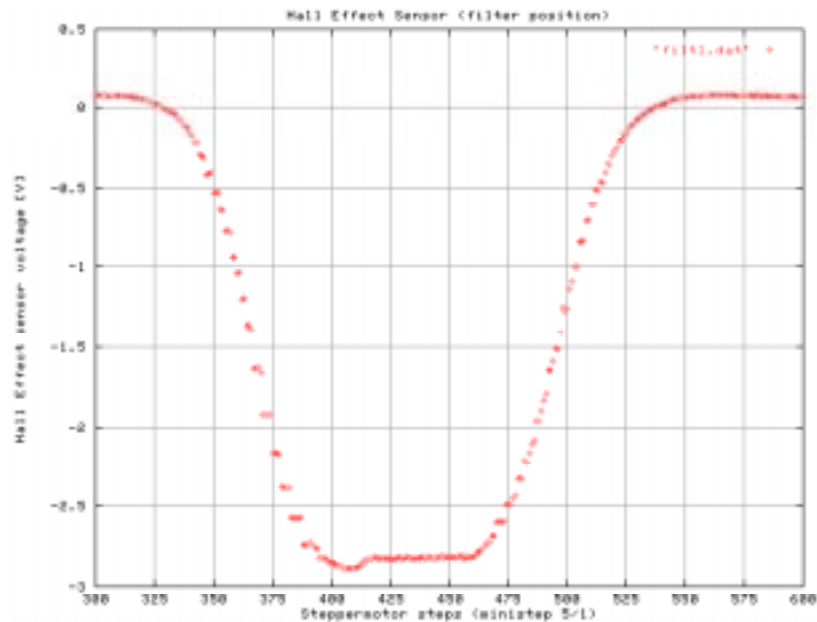
3.1.1.2 Sensor data plots (Confirmation magnet)

Figure 5 is a data plot of the Hall-Effect sensor values near a datum magnet. Note that the (negative) peak is flattened. This demonstrates the effectiveness of the Geneva mechanism which is used to turn the filter wheel. (The Geneva mechanism causes the motion of the wheel to be non-uniform. The motion of the wheel is slowed in the vicinity of the filters, making the hardware relatively insensitive to positioning errors.) The data values become positive in the wings of the peak. This is probably a real effect, due to the reversal of the direction of the magnetic field near a magnet.

This data was taken using a 5/1 mini-step stepper motor controller. In practice, the mini-steps may prove to be useless, because the motor will tend to fall into a detente when power to the motor is removed.

FIGURE 6.

Hall effect sensor readings (Filter magnet)



3.1.2 Continuous-positioning mechanisms (OIWFS focusing stage)

In some cases, a single sensor will be unable to provide adequate resolution. Higher resolution will be provided by using a combination of magnets and sensors on different parts of the mechanism (e.g., for the focusing stage, one magnet/sensor pair on the moving part of the stage, and a second pair on the main gear).

Figure 7 describes an (over)simplified model of a focusing stage, which requires two sensor pairs to define the datum position. The coarse sensor is able to position the stage in the vicinity of the datum, unambiguously, but with poor resolution. The gear sensor is able to determine the position very accurately, but the true position can be incorrect by an integral number of gear rotations.

A dual-magnet linear sensor arrangement is shown. The coarse-positioning or “linear” magnets are placed with opposite polarities, producing two adjacent peaks, one positive, and one negative. Between the peaks, there is a ramp, which is (almost) linear. The stage is designed so that it always operates in the linear region, providing a useful (though not extremely accurate) encoding of the position of the stage. Note that any changes in the sensitivity of the sensor electronics or constant offset can affect this encoding scheme. This is sufficient, because the encoding scheme needs only to position the stage at the datum, to within one turn of the main gear.

FIGURE 7. A two-sensor arrangement (OIWFS Focusing stage)

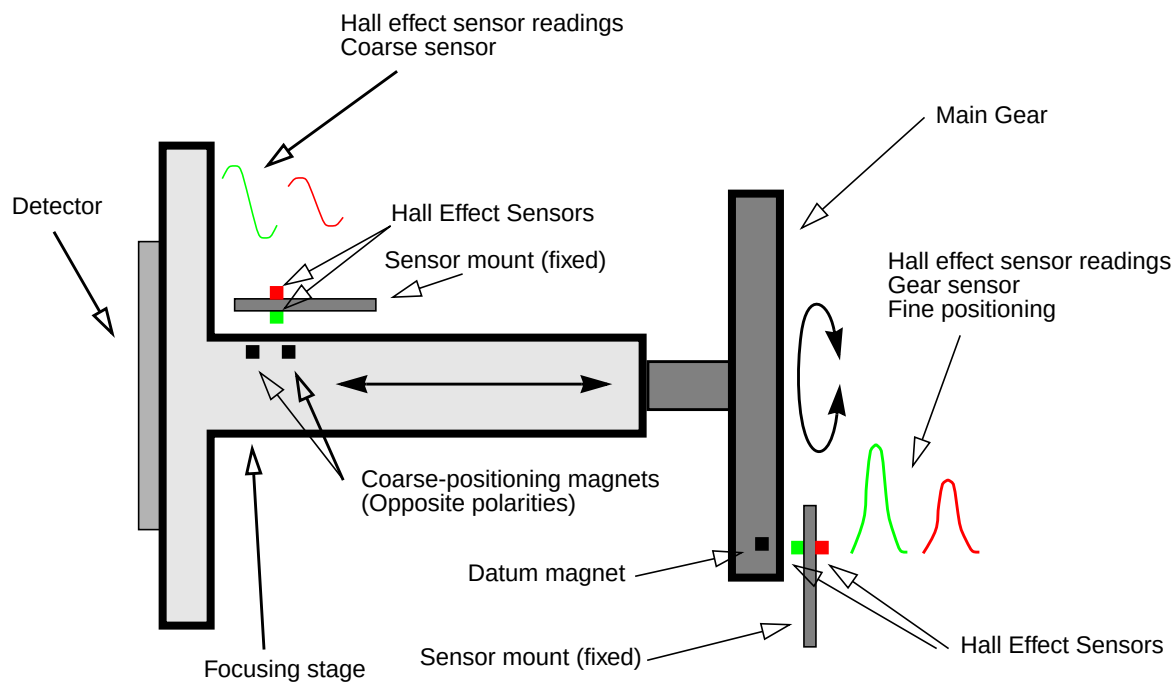
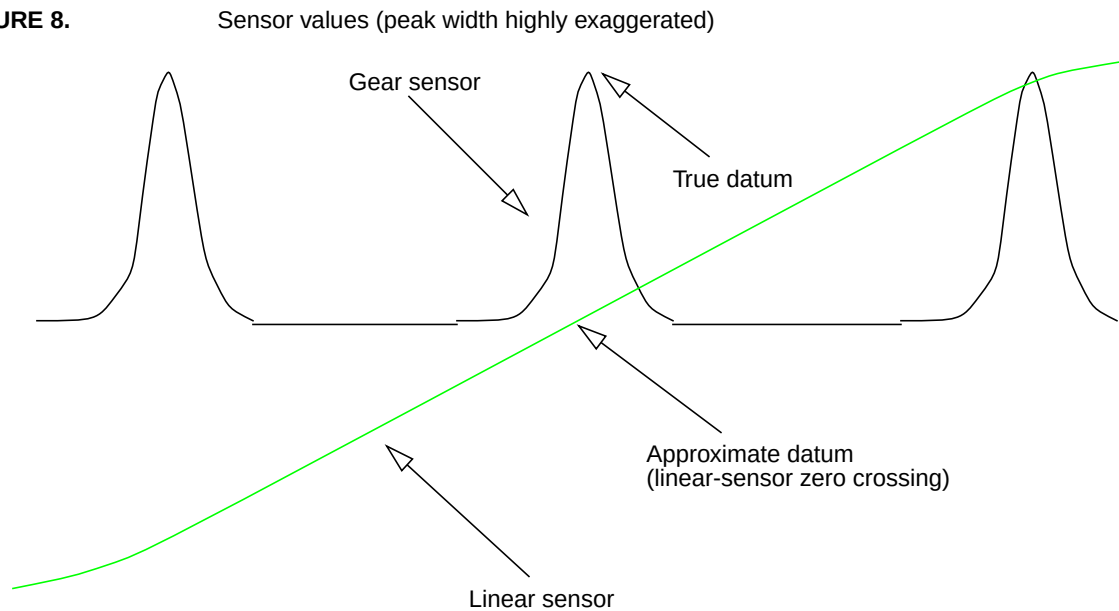


Figure 8 shows the expected reading from the Hall-effect sensors.

FIGURE 8.



Unlike the filter wheel, the Hall-effect sensors here are mounted on top of each other. This arrangement is simpler than for the filter wheel, because it requires only a single magnet, not a pair of magnets. Unfortunately, because it is not possible to insure that the sensors are precisely concentric with each other, there is a slight difference in the peak position. This must be compensated for via an offset value, which must be determined empirically. Furthermore, the signal strength for the sensor which is further away from the magnets is noticeably weaker.

One sensor is designated as the “primary” sensor. This sensor will be connected to the “home” input of the Oregon Microsystems stepper motor controller, which will speed up the datuming process. If this sensor fails, then the datum position may still be found, but will require a slower search process.

3.1.2.1 Reference points

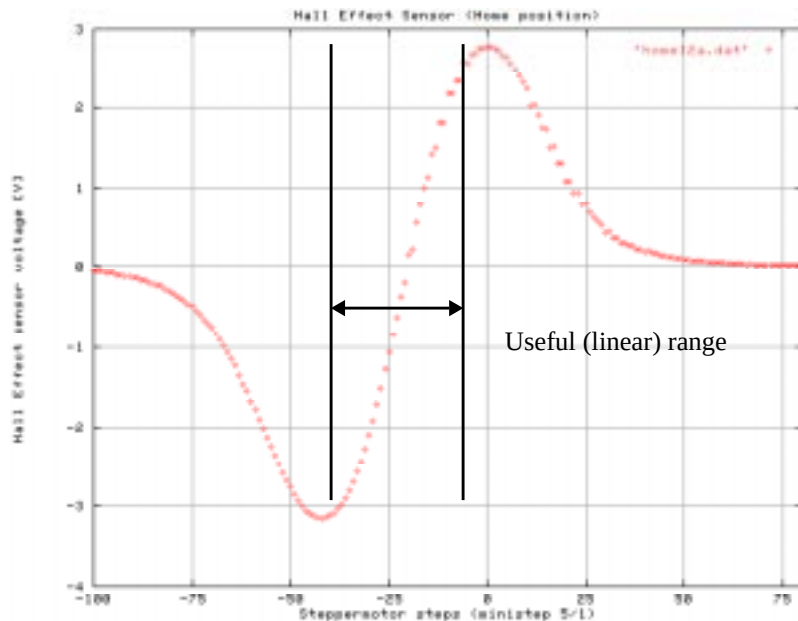
Although there are no predefined locations, a list of (arbitrarily chosen) reference points will be defined. To insure proper operation of the system, a command (“DIAGNOSE”) will be provided to move the mechanism to each reference point, and the expected value of the hall-effect sensors will be compared the actual value.

3.1.2.2 Sensor data plots (“linear” arrangement)

Figure 9 shows measured values for the Hall effect sensors in a “linear” arrangement. Note that the data values are almost linear in the region between the peaks. This provides a useful partial encoding scheme; however it is too susceptible to instabilities in the electronics to be used as a precise encoding scheme.

FIGURE 9.

Hall effect sensor readings (“linear” magnet arrangement)



3.1.3 Other continuous-positioning mechanisms (science focus and gimbals)

Unfortunately, not all of the continuous-positioning mechanisms are identical. The focussing stage for the science detector and the OIWFS gimbal mirrors are similar to the focussing stage for the OIWFS detector, but are not identical.

3.1.3.1 The focussing stage for the science detector

The larger range of travel of the focussing stage for the science detector means that the full range of travel of the stage cannot be uniquely encoded, as it is for the OIWFS focussing stage. To get around this problem, additional magnets are installed on the translation stage to insure that there is a region in which the signal value is a one-to-one function of the position. The ends do not provide a unique signal, but the sign of the signal is sufficient to indicate the location of the linear region.

FIGURE 10. A two-sensor arrangement (Science Focusing stage)

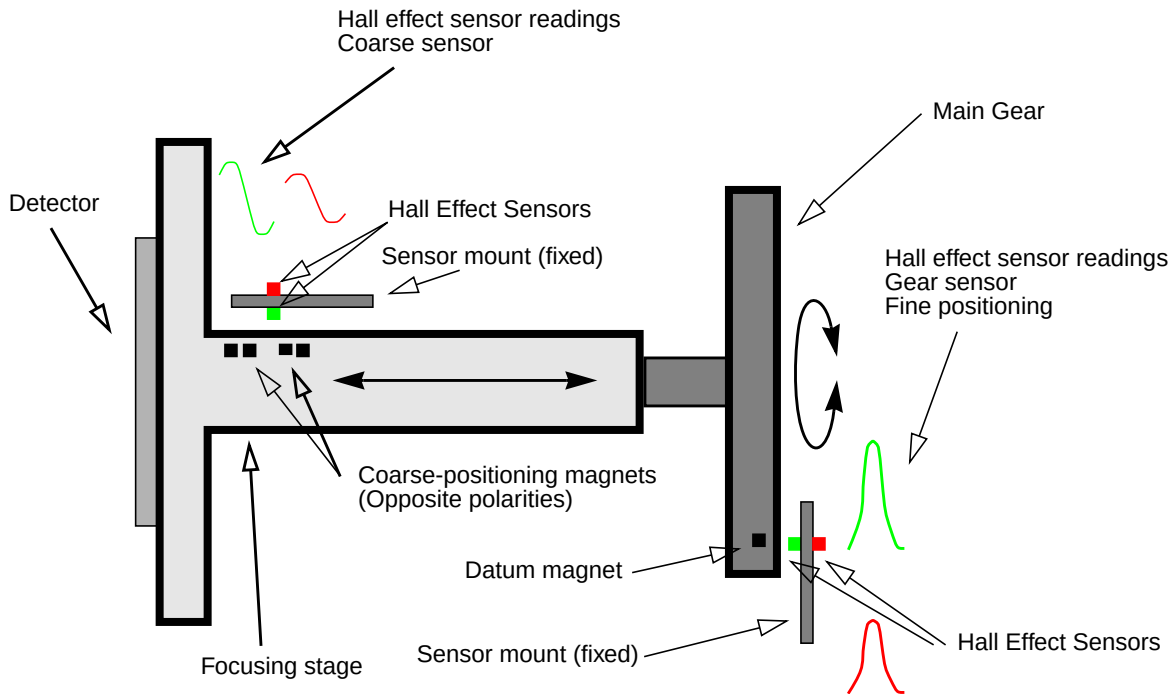
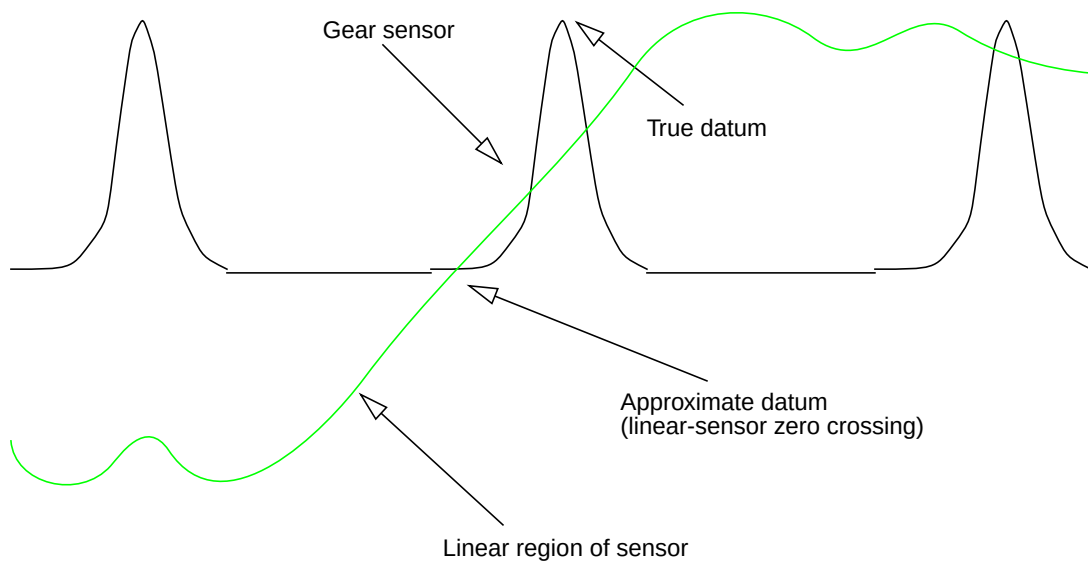


Figure 11 shows the expected reading from the Hall-effect sensors.

FIGURE 11. Sensor values (peak width highly exaggerated)



3.1.3.2 The OIWFS gimbal mirrors

The larger range of travel of the gimbal mirrors stage and the precise positioning requirements require still another variation for the gimbal mirrors. Instead of a linear positioning sensor, a single magnet is used, and the location of the peak is used to determine the true home position.

FIGURE 12. A two-sensor arrangement (gimbal mirror stage)

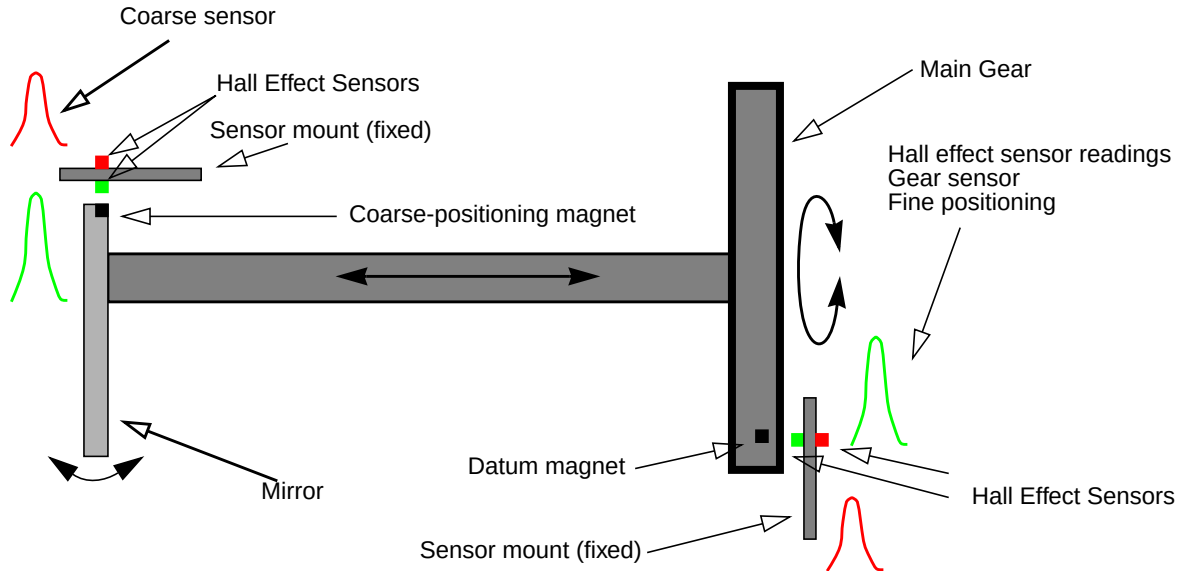
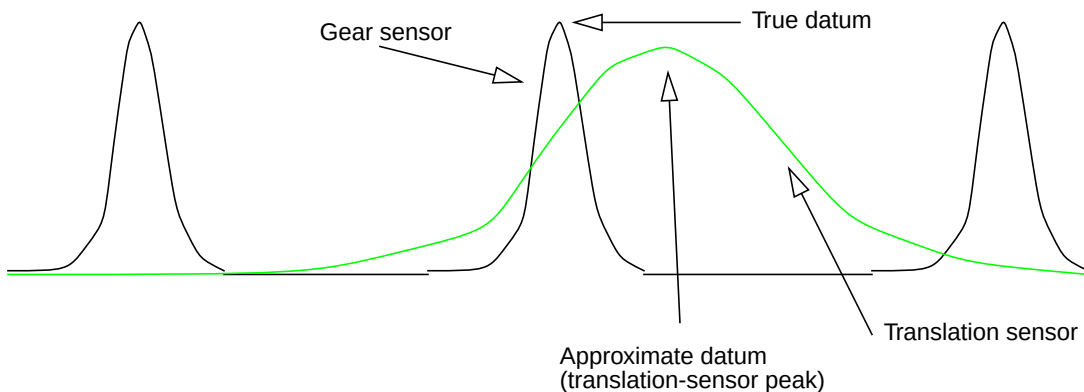


Figure 13 shows the expected reading from the Hall-effect sensors.

FIGURE 13. Sensor values (peak width highly exaggerated)



3.1.4 Two position mechanisms (the pupil viewer)

The pupil viewer has two defined positions. Each position is located at a hard stop, at an end of travel.

FIGURE 14. A one-sensor arrangement (pupil viewer))

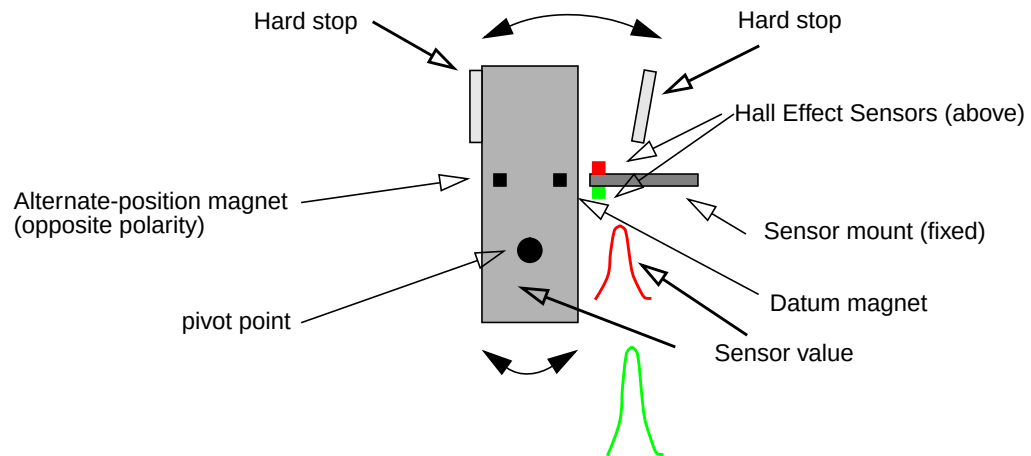
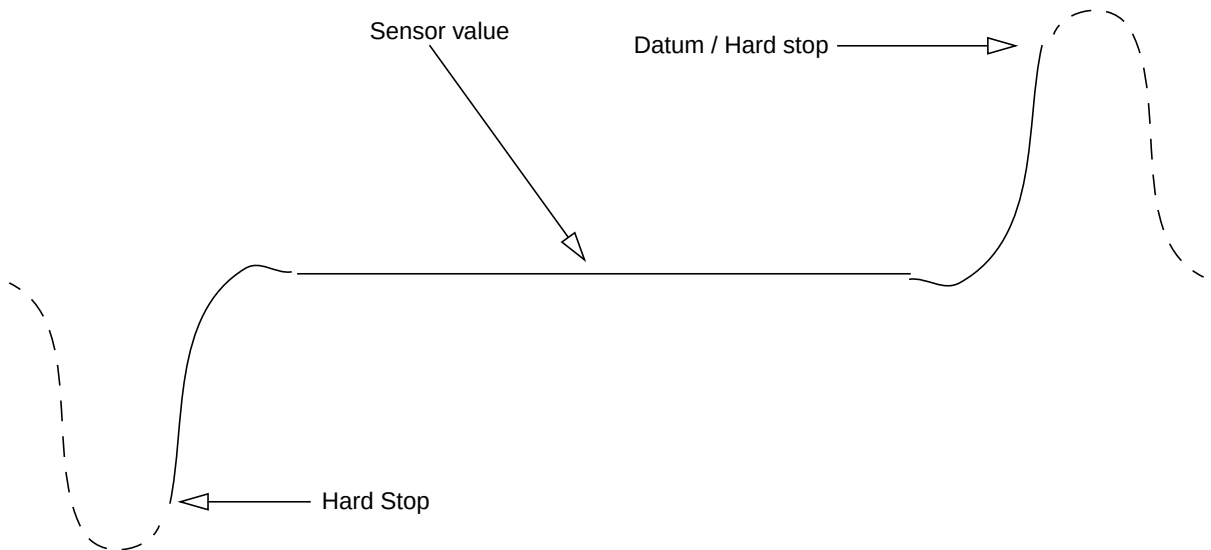


Figure 13 shows the expected reading from the Hall-effect sensors.

FIGURE 15. Sensor values (peak width highly exaggerated)



3.2 Peak Finding Algorithm³

For the rest of this document, we will refer to “finding the peak value” associated with each magnet/sensor pair, using the following algorithm

Locate the largest sensor value with a brute-force search.
Sample the sensors values near the maximum sensor value.
Fit a parabola to the sensor values.
Use the peak of the parabola as the sensor value.

This symmetric algorithm should be relatively insensitive to slight changes in the gain of the electronics, and to any biases in the electronics.

3.3 Stepper-motor errors

The hardware will be designed so that the stepper motors will never lose counts if all mechanisms are working properly. If any counts are lost, a problem exists which must be detected and corrected. The Hall-effect sensors may be able to detect the problem. If the sensors are able to detect a problem, an ERROR will be generated so that appropriate error recovery may be taken.

3.4 Backup sensors

All of the Hall-effect sensors have backup sensors. The system will be designed to use a primary sensor to datum the corresponding mechanism and to confirm the instrument configuration, and a secondary sensor for confirmation purposes only. Either sensor may be disabled if it fails. If either sensor is disabled, it may not be possible to confirm the configuration of the corresponding mechanism. If the primary sensor fails a slower datuming algorithm will be used. If both sensors are disabled, it will not be possible to confirm the current position nor will it be possible to datum the mechanism. Disabling either sensor will generate a WARNING.

3.5 Hard Limits and Limit Switches

Most of the mechanisms do not have hard limits and can be driven continuously indefinitely. The exceptions are described in the following section.

3.5.1 The focusing stages

The wavefront-sensor focusing stage and the science-detector focusing stage both have hard stops. Neither mechanism has a limit switch or a passive way of detecting that the mechanism has hit a limit. Both mechanisms are designed so that they will not be damaged by *occasionally* hitting a hard stop. The software is designed with soft limits and it should be impossible to hit a hard stop unless there is a hardware problem. If a hard stop is encountered, it will be necessary to redatum the mechanism.

3.5.2 The Pupil Viewer

3. The peak-finding algorithm is implemented within the HallStep record, documented in [35] in Section 4 on page 6.

The pupil viewer is designed so that it always hits a hard stop whenever it hits the limits of its travel.

3.5.3 The OIWFS Gimbal Mirror

The gimbal mirror can be damaged by overrunning its limits. There are limit switches to prevent this from happening. These switches are not intended to be used in normal operation. The software will be designed with soft limits so that the mechanism will never hit a limit.

If a motion command causes the hardware to hit a limit switch, the mechanism will stop and the motion command will complete with an error message. It will then be possible only to back the mechanism out of the limit. It should not be necessary to redatum the mechanism after hitting a limit.

Warning: The limit switches for both axes are tied together. If either axis hit the positive limit, then both axes will stop, and it will only be possible to move the axes in a negative direction. The same holds true for the negative limits. It is possible to reach a state in which it is not possible to move the mechanism: If one axis actuates the positive limit, and the other axis actuates the negative limit, then the mechanism will not be able to move. In this case, it will be necessary to determine which mechanism is in which limit, and to carefully back out of each limit in the proper direction. The procedure for determining which limit switch has been actuated by which axis is described in the hardware reference manual.

3.6 Simultaneous motor motion

It will not be possible to move all motors in the system simultaneously, because of power restrictions. (This will be handled by the lower level software, and should be transparent to the upper level interface.)

Whenever a motor becomes active, a counter will be incremented. When the maximum number of motors is running, new operations will be delayed until the motor count drops below the maximum.

3.7 Stepper Motor Control Module Failure

If the motor draws too much current, the stepper-motor control modules can enter into a fault state. The motor is unable to operate if a fault occurs. This fault generates an immediate ERROR until the stepper-motor control module is reset. A command will be provided to reset the fault, which should be a very infrequent event.

Note: A fault condition also exists if power is not applied to the control module. This is normal. Obviously, it is impossible to reset this fault. It is not possible to distinguish this from the over-current fault condition.

3.8 Diagnose

A DIAGNOSE command will be provided for each component (This should not be confused with the standard TEST and VERIFY commands.) This command will move the hardware and monitor the Hall-effect sensors in order to determine if the hardware is functioning. It should be possible to discover a single, failed, sensor. It will not be possible to distinguish a

double sensor failure from a failed motor. Unlike the TEST command, the DIAGNOSE command *will* move the hardware.

3.9 Redatum

A REDATUM command will be provided for each component. (This should not be confused with the standard TEST and VERIFY commands.) This command will move to the expected datum position and confirm that it really is at the datum. If the verification fails, then an error will be returned. This command will not be used as part of the normal configuration-change procedure. It will only be used if a possible hardware problem is suspected.

4 Environment Control

4.1 Cooling Motor

The closed-cycle coolers are driven by the same stepper-motor controller as the niri mechanisms.

4.2 Temperature regulation

The NIRI and NIRI OIWFS temperature control is handled with three off-the-shelf Omega temperature controllers. These controllers accept ascii commands via their serial ports. A custom EPICS record is used to pass commands to these temperature controllers.

4.3 Heating Resistors

Warm-up is handled by a combination of heating resistors, which can be disabled using the digital input/output board (controlled using an EPICS binary output record) and the Omega temperature controller.

It will be necessary to coordinate the warming of the components with the warming of the detector, to insure that the detector is always slightly warmer than the surrounding instrument. Overall control of the cooling motors and the temperature regulation is handled by a custom record.

4.4 Temperature Sensors

An undetermined number of temperature sensors will monitor the behavior of the system if the temperature is out of the valid range an ERROR will be generated.

The temperature sensors will be read using an off-the-shelf Omega digital thermometer, which accepts commands via a serial port. A custom record has been provided to read the temperature.

5 Special Calibration Programs

There will probably be a need for special purpose software for engineering purposes, e.g., to generate the look-up tables of filter positions. These programs will never be used in normal operation. After delivery of the instrument to gemini, the programs will only need to be run after daytime engineering in which significant changes are made to the NIRI hardware.

These programs will be documented in [5].

6 The Gemini Interlock System

The appropriate hardware and software support will be provided for the Gemini Interlock System, but NIRI will never generate an interlock event, and will need to carry out no actions in response to an interlock demand.

7 Special Epics Records

7.1 Modified records

7.1.1 Motor record (OMS VME8, VME44 device/driver)

The EPICS motor and OMS VME8, VME44 device provides support for the Oregon Micro-systems VME8-8 board. We have produced a modified motor record and device support. (If necessary to avoid conflicts, this record will be called the “nirimotor” record.

- The record is modified to allow direct control over the AUX output. (The AUX output is used to control the power to the stepper motors from the stepper motor driver modules.) The PWR field has three possible values: “ON”, “OFF”, “AUTO”. “AUTO” is the default and is backwards compatible with the existing record.
- Constants and global variables have been renamed, to have a consistent “MOTOR_” or “motor” prefix.

7.2 Custom Records, Device Support, and Subroutine Records

The following will probably be necessary. They will be some combination of custom records, new device support for existing records, subroutine records, gensub records, and SNL code:

- Records/code to control the cooling and warming of the instrument;
- Records/code to optimize the motions of the filter wheels;
- Records/code to control the cooling motor;
- Records/code to control the temperature controller.

